QUAID-I-AZAM UNIVERSITY ISLAMABAD

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DETECTION OF CLICKBAIT THROUGH MACHINE LEARNING

SOFTWARE PROJECT TECHNICAL DOCUMENTATION

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Abstract

Detection of clickbait through Machine Learning

Due to the fast development of social media and information websites, misleading and exaggerated news titles, have become increasingly more commonplace. These misleading information titles goal to seize interest of the reader and maximize the quantity of clicks. However, they'll mislead the reader and create false expectations. In this assignment, the development of a **Clickbait Detection System** based totally on system getting to know may be discussed. The system might be capable to analyze news titles or small information articles to decide whether they're clickbait or valid information.

To increase the **Clickbait Detection System, Natural Language Processing (NLP)** strategies might be hired to procedure the input information, extract essential features, and train a class version to become aware of clickbait, The system might be capable of be given news titles or small news articles as input and offer an immediate prediction with a self assurance level. The machine will also be capable of pick out the vital phrases or terms that inspired the prediction to help the person better recognize the purpose behind the prediction. In addition to this , the system can also be able to maintain a history of the predictions and offer accuracy, precision, and take into account metrices to determine the performance of the type version.

By integrating gadget mastering with the prediction mechanisms, the proposed Clickbait Detection System may be an sensible machine to become aware of misleading news titles. The proposed device could be beneficial for customers to critically examine the virtual content they eat, and the overall recognition of deceptive news titles can be elevated

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Chapter No 4

Software Testing

Introduction

In this document we will write Acceptance test to test our project.

Acceptance Test for UC-1 User Registration

Test ID	TC-1	
Description	To test that the Clickbait Detection System lets a new consumer create an account efficiently and successfully handles incorrect or invalid registration information.	
Setup	1. Ensure the Clickbait Detection System is running in the browser. 2. Ensure the test email is not already registered in the database.	
Instruction	Valid Input	1. Open the system 2. Navigate to the User Registration page. 3. Enter the following details: 1. Name: Test User 2. Email: testuser@example.com 3. Password: Test@123 4. Confirm Password: Test@123 4. Click the Register button.
	Invalid Input	1. Enter invalid email: testuser@invalid 2. Enter mismatched passwords: Test@123 and Test@476 3. Enter already registered email: existing@example.com 4. Click the Register button for each case.
Expected Result	Valid Input	• System validates all fields. • New user account is created in the database. • System presentations message: “Registration Successful”. • User is redirected to the Login Page.
	Invalid Input	1. For invalid e-mail: System displays “Invalid e-mail format”. 2. For mismatched passwords: System presentations “Password does not fit”. 3. For current email: System shows: “Email already registered”. 4. No account is created in invalid case.

Table 14 Acceptance Test for UC-1 User Registration

Acceptance Test for UC-2 User Login

Test ID	TC-2
Description	To make sure a registered consumer can log in to the Clickbait Detection System and that the system blocks invalid login attempts.
Setup	1. Ensure the Clickbait Detection System is running in the browser. 2. Make sure that valid user account is already exists in the database. • Email: testuser@example.com • Password: Test@123

Instruction	Valid Input	1. Open the system 2. Navigate to the Login page. 3. Enter the credentials as following: • Email: testuser@example.com • Password: Test@123 4. Click the Login button.
	Invalid Input	1. Enter wrong Password: Test@ew43 2. Enter e mail that isn't always even registered: gty@exg.Com 3. Let the sector of password and electronic mail empty. 4. Try to click login button over and over.
Expected Result	Valid Input	• System verifies the credentials from the database. • User is efficaciously authenticated. • System redirects the user to the Dashboard / Home web page. • A fulfillment message may be displayed (e.g., "Login Successful").
	Invalid Input	1. For wrong password: System shows: "Invalid electronic mail or password". 2. For unregistered e mail: System presentations: "Account does no longer exist". 3. For empty fields: System displays: "Please fill all fields". 4. User isn't always allowed to get admission to the system.

Table 15 Acceptance Test for UC-2 User Login

Acceptance Test for UC-3 Reset Password

Test ID	TC-3	
Description	To verify that a registered user can efficiently reset their password and that the gadget properly handles invalid reset tries.	
Setup	1. Make sure Clickbait Detection System is running. 2. Ensure that user account already exists in the database: • Email: testuser@example.com • Old Password: Test@123	
Instruction	Valid Input	1. Open the machine. 2. Navigate to reset password web page. 3. Enter the registered e mail: testuser@instance.Com. 4. Enter new password: Newpass@123 5. Confirm new password: Newpass@123 6. Click the Reset password button.
	Invalid Input	1. Enter unregistered electronic mail: unknown@example.Com 2. Enter mismatched passwords: Newpass@123 and Newpass@671

		3. Leave fields empty. 4. Click the Reset password button for each case.
Expected Result	Valid Input	• System verifies the email exists. • Password is up to date in the database. • System displays: “Password Reset Successful”. • User can log in the use of the new password.
	Invalid Input	1. For unregistered e mail: System displays: “Email not found”. 2. For mismatched passwords: System presentations: “Passwords do not match”. 3. For empty fields: machine displays: “Please fill in all required fields”. 4. Password isn't always modified in invalid cases.

Table 16 Acceptance Test for UC-3 Reset password

Acceptance Test for UC-4 User Logout

Test ID	TC-4	
Description	To validate that an authentication consumer can securely log off of the Clickbait Detection System and that the session is properly terminated.	
Setup	1. Ensure the Clickbait Detection System is running inside the browser. 2. Ensure the user is already logged into the gadget.	
Instruction	Valid Input	1. While logged in, navigate to the dashboard or home page. 2. Click at the Logout button.
	Invalid Input	1. Attempt to get entry to a covered page (e.G., Dashboard) after logging out. 2. Use the browser again button after logout to test consultation behavior.
Expected Result	Valid Input	• The system terminates the lively person consultation. • The user is redirected to the Login page. • A confirmation message such as “Logged out correctly” can be displayed.
	Invalid Input	1. After logout, the consumer must no longer be able to access blanketed pages. 2. If attempting to get admission to a secured page, the machine redirect the consumer to the Login page. 3. The preceding consultation need to now not stay active.

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Acceptance Test for UC-5 Input Headlines / News Text

Test ID	TC-5

Description	To make certain that a user can input and publish a headline or news textual content for clickbait evaluation, and that the device validates the enter correctly before processing it.	
Setup	1. Ensure the Clickbait Detection System is going for walks. 2. Ensure the person is logged into the machine. 3. Ensure the text input interface is offered from the dashboard.	
Instruction	Valid Input	1. Navigate to the textual content enter segment. 2. Enter a legitimate headlines or short news text, for instance: 3. “You Won’t Believe What Happened Next!”. 4. Click the Analyze or Submit button.
	Invalid Input	1. Leave the text enter discipline empty. 2. Enter handiest special characters (e.G., @@@@). 3. Enter extraordinarily brief meaningless text (e.g., “aa”). 4. Click the Analyze button for each case.
Expected Result	Valid Input	• The device accepts the enter text. • The text is proven correctly. • The machine forwards the textual content to the system studying version for prediction. • No validation errors are displayed.
	Invalid Input	1. If the sphere is empty: System shows: “Text field can't be empty”. 2. If the input is invalid or meaningless: System displays: “Please enter legitimate headlines or news text”. 3. The gadget does not proceed to prediction in invalid cases.

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Acceptance Test for UC-6 Upload Dataset

Test ID	TC-6	
Description	To make certain that the machine permits customers to upload a CSV dataset for schooling the system gaining knowledge of model and efficaciously handles invalid dataset uploads.	
Setup	1. Ensure the Clickbait Detection System is strolling. 2. Ensure the consumer is logged into the gadget. 3. Ensure the dataset upload interface is accessible from the dashboard. 4. Prepare a legitimate dataset record in CSV format containing headline text and labels (clickbait / non-clickbait).	
Instruction	Valid Input	1. Navigate to the Dataset Upload phase. 2. Click the Upload Dataset button. 3. Select a valid CSV report containing clickbait dataset statistics. 4. Click the Upload / Submit button.

	Invalid Input	1. Attempt to upload a record that isn't in CSV layout (e.g., .Txt or .Pdf). 2. Upload a CSV report with lacking or wrong columns. 3. Attempt to add an empty report. 4. Click the Upload button for each case.
Expected Result	Valid Input	• The machine successfully uploads the dataset file. • The dataset is stored in the gadget for similarly processing and model schooling. • A affirmation message which include "Dataset uploaded effectively" is displayed.
	Invalid Input	1. If record layout is not CSV: System displays: "Invalid record format. Please upload a CSV file". 2. If dataset structure is wrong: System shows: "Dataset layout is invalid". 3. If report is empty: System displays: "Uploaded document carries no information". 4. The dataset isn't always popular in invalid instances.

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